

THE MAGIC OF LENSES

How They Bend Light and Make Things Bigger!

Key Words to Remember:

- **Lens:** A clear, curved piece of glass or plastic that bends light.
- **Transparent:** Allows light to pass through clearly.
- **Refraction:** The bending of light when it passes through a transparent object.

1. Meet the Lenses!

1. Convex Lens

"The Magnifier"

- **Shape:** Thicker in the middle (like a *Laddoo!*).
- **Action:** Bends light rays together (**converges**).
- **Result:** Makes objects look **BIGGER!**
- **Uses:** School microscopes, magnifying glasses, mobile camera lenses, reading glasses.



Memory Trick:

"Convex Comes OUT & Converges — Convex = BIG!"

2. Concave Lens

"The Shrinker"

- **Shape:** Thinner in the middle, caves inward (like a bowl!).
- **Action:** Bends light rays apart (**diverges**).
- **Result:** Makes objects look **SMALLER!**
- **Uses:** Myopia (near-sightedness) glasses, door peepholes, flashlights.



Memory Trick:

"Concave CAVES IN like a bowl — Concave = SMALLER!"

2. Lens Detective Activity!

Can you identify which type of lens is used in these everyday objects?

Everyday Object	Lens Type	What It Does
 Magnifying Glass	Convex Lens	Makes tiny bugs and text bigger!

 Student's Myopia Glasses	Concave Lens	Helps see distant objects clearly!
 Mobile Camera Lens	Convex Lens	Focuses light to take sharp photos!
 Plain Window Glass	No Lens	Light passes straight without bending.

☀ Quick Recap:

- Lenses bend light using a process called **refraction**.
- **Convex lenses** bring light together (converge) and make things look larger.
- **Concave lenses** spread light apart (diverge) and make things look smaller.



Practice Worksheet: Test Your Knowledge!

Answer the following questions to become a certified Lens Detective!

Question 1 (Fill in the blank):

A clear, curved piece of glass or plastic that bends light is called a _____.

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Question 2 (Multiple Choice):

What is the bending of light called when it passes through a lens?

- a) Reflection b) Refraction c) Shadow
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Question 3 (True or False):

A convex lens is thinner in the middle and thicker at the edges. (True / False)

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Question 4 (Short Answer):

Which type of lens brings light rays together (converges light)?

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Question 5 (Short Answer):

Does a magnifying glass use a convex lens or a concave lens?

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Question 6 (Fill in the blank):

A concave lens makes objects appear _____ in size.

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Question 7 (Multiple Choice):

What shape is a convex lens compared to in the lesson?

- a) A bowl b) A laddoo c) A flat plate
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Question 8 (Short Answer):

Name one device used by scientists or photographers that contains a convex lens.

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Question 9 (Short Answer):

What kind of lens is used in myopia glasses to help kids see the blackboard clearly?

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Question 10 (True or False):

Window glass is a lens because light passes through it. (True / False)

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Question 11 (Matching):

Match the lens term to its memory trick:

- 1) Concave Lens → (_____) A. Convex Comes OUT & Converges
2) Convex Lens → (_____) B. Caves IN like a bowl
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Question 12 (Fun Drawing / Application Question):

Imagine you are a detective examining a tiny ant. Which lens would you choose to make the ant look bigger, and why?

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